

Geometry Guided Generalizable-FER

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Facial Emotion Recognition(FER)

What is the task

- Detect emotion from an image(classification)
 - Input: an image of a human face
 - Output: label from 7 emotions
(happy, surprise, anger, sad, fear, disgust, neutral)



Facial Emotion Recognition(FER)

Generalizable Facial Expression Recognition

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- Computer Vision--ECCV 2024
- Data: RAF-DB, FERPlus, AffectNet, SFEW2.0, MMA

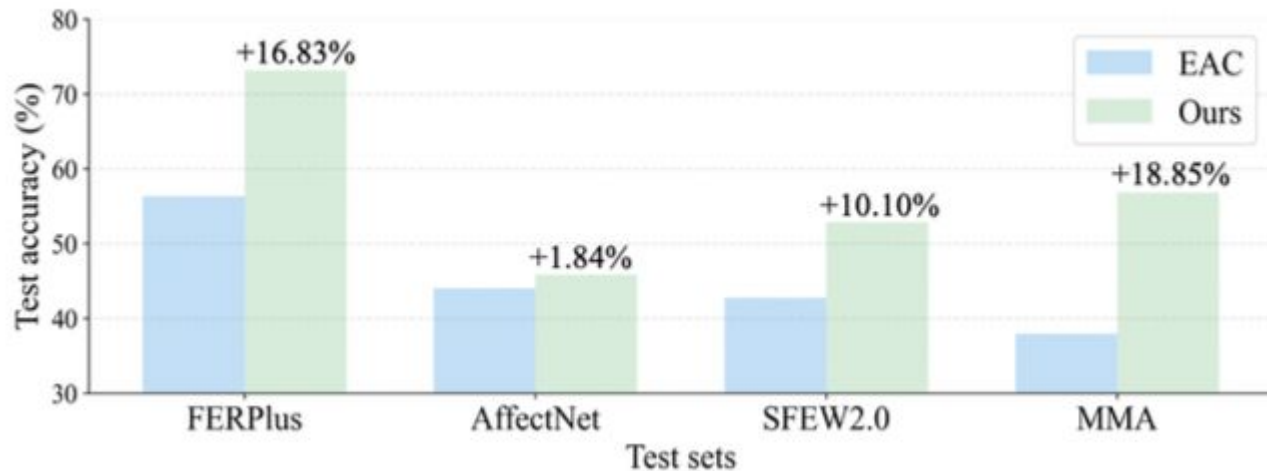
About the Dataset

Dataset	Data Points	Description
RAF-DB		- 7 basic expressions by 40 trained human coders. - surprise, fear, disgust, happiness, sadness, anger, neutral.
FERPlus		- FER2013 with cleaner labels collected by the Google search engine. - Only utilize the same 7 basic expressions.
AffectNet		- Contains 8 expressions (7 basic expressions and contempt). - Only utilize the same 7 basic expressions.

About the Dataset

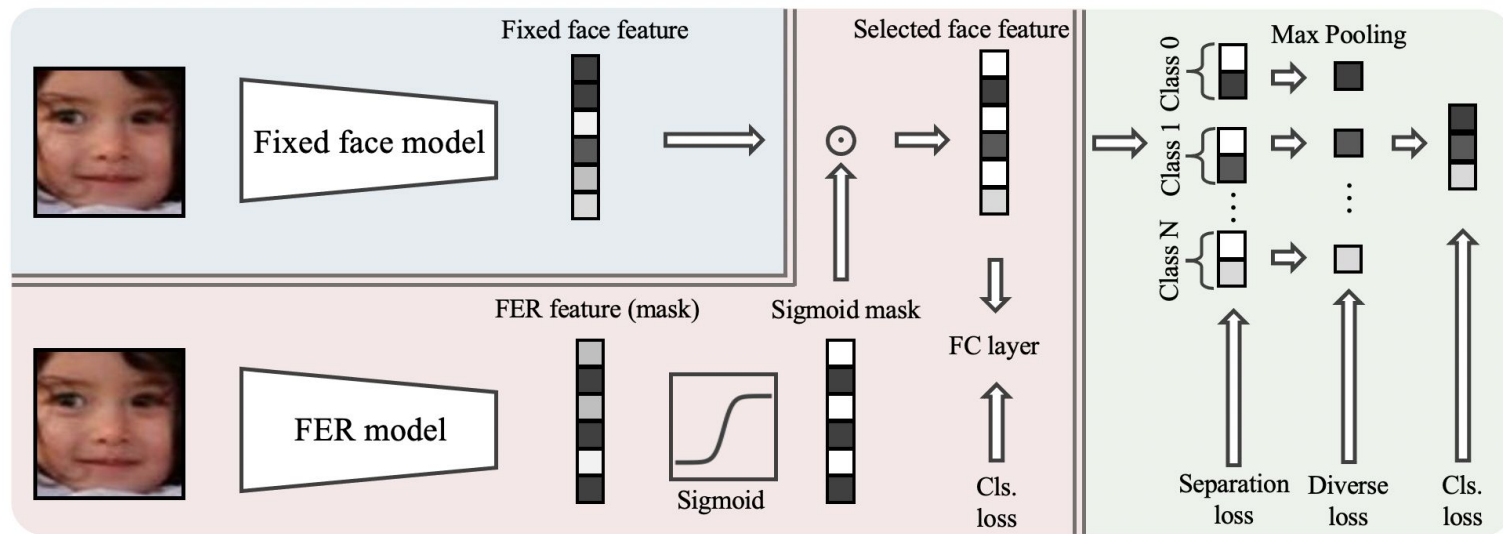
Dataset	Data Points	Description
SFEW2.0		- Most commonly used version of SFEW - Each image assigned to one of 7 basic expressions
MMA		- Majority of expressions from individuals of European and American descent. - Each image assigned to one of 7 basic expressions

Generalizable-FER



- Problem:
 - Standard SOTA models perform well on training sets but fail when tested on unseen datasets due to domain gaps.
 - Overfitting: Models learn domain-specific noise (lighting, background) instead of general emotion.

Generalizable-FER



- Step 1: Extract fixed face features via large pre-trained models (CLIP).
- Step 2: Learn a Sigmoid Mask to select only expression-relevant channels
- Step 3: Classify based only on selected features to avoid noise.

$$l_{train} = l_{cls} + \lambda l_{sep} + \beta l_{div}.$$

$$l_{div} = 1 - \frac{1}{N_c} \sum_{i=1}^N \sum_{j=1}^L \tilde{\mathbf{F}}_{max_{ij}}$$

$$l_{sep} = -\frac{1}{N} \sum_{i=1}^N \left(\log \frac{e^{\mathbf{F}_{y_i}^d}}{\sum_j^L e^{\mathbf{F}_j^d}} \right)$$

Generalizable-FER

Method	RAF-DB	FERPlus	AffectNet	SFEW2.0	MMA	Mean
SCN (CVPR2020)	87.32	<u>58.37</u>	42.85	44.89	36.52	53.99
RUL (NeurIPS2021)	88.66	57.89	43.82	<u>46.91</u>	37.11	54.88
EAC (ECCV2022)	89.54	54.38	<u>43.91</u>	43.39	<u>37.27</u>	53.70
OFER (ICCV2023)	89.07	53.90	42.73	43.88	36.43	53.20
CAFE	88.72	73.16	45.86	52.86	56.80	63.48

Method	FERPlus	RAF-DB	AffectNet	SFEW2.0	MMA	Mean
SCN (CVPR2020)	86.80	<u>68.71</u>	32.42	43.10	59.12	58.03
RUL (NeurIPS2021)	88.40	51.89	35.88	<u>45.90</u>	58.00	56.01
EAC (ECCV2022)	89.03	58.62	<u>36.49</u>	45.79	<u>59.89</u>	57.96
OFER (ICCV2023)	89.26	55.67	35.84	44.52	59.37	56.93
CAFE	89.51	72.91	39.44	49.38	60.14	62.28

- more generalized but not enough to capture fully generalizable facial expressions

Limitation

- Geometric Blindness

Vision encoders struggle to capture fine-grained facial structure information (landmarks) and precise spatial relationships essential for subtle expressions. (Multimodal Prompt Alignment for Facial Expression Recognition, IEEE, 2025)

- Identity Entanglement

Facial Expression Analysis (FEA) suffers from severe identity bias, where expressions are entangled with task-irrelevant factors like identity and head pose. (Norface: Improving Facial Expression Analysis by Identity Normalization, ECCV, 2024)



Need a specific facial expression reference!

Goal

- Goal: Improve cross-dataset generalization of the CAFE model
- Setup:
 - Train on a single dataset
 - Evaluate on 4 unseen datasets
- Proposed Improvements:
 - AU-Guided Mask
 - Class Imbalance Loss
 - Data Augmentation
- Validation: Ablation study to analyze individual and combined effects (Accuracy)

Generalizable-FER

Improvement

1. Masking

Current Model's Masking Problem :

- **Learn** the feature **blindly** based on sigmoid
- **Lacks guidance** causes the model to miss important face features

Our Improvement:

Action Units (AU)-Guided Masking

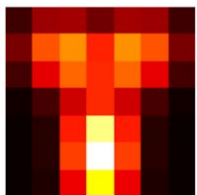
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Action Units (AU)

- Codes from the **Facial Action Coding System (FACS)** by Ekman & Friesen (1978)
- Each AU represents a specific facial muscle movement

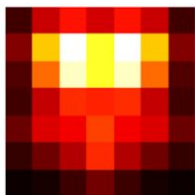
“Every facial expression is a combination of specific AUs”

7x7 AU Spatial Heatmap



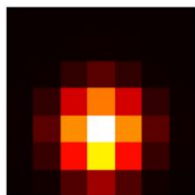
Surprise

Brows
+
Jaw Dropping



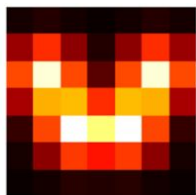
Fear

Brows
+
Eyes + Mouth



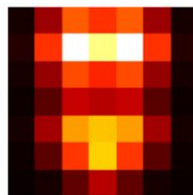
Disgust

Nose
+
Mouth



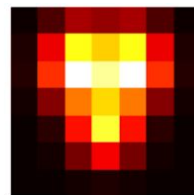
Happy

Cheeks
+
Lip corners



Sad

Brows
+
Lip + Chin



Angry

Brows lowered
+
Lip pressed

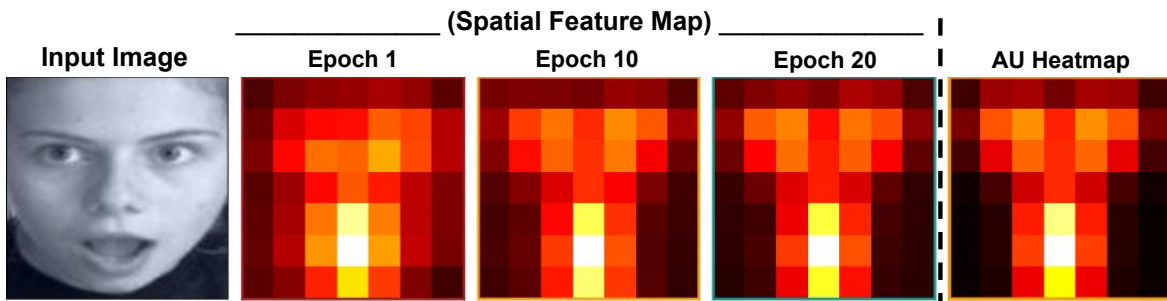
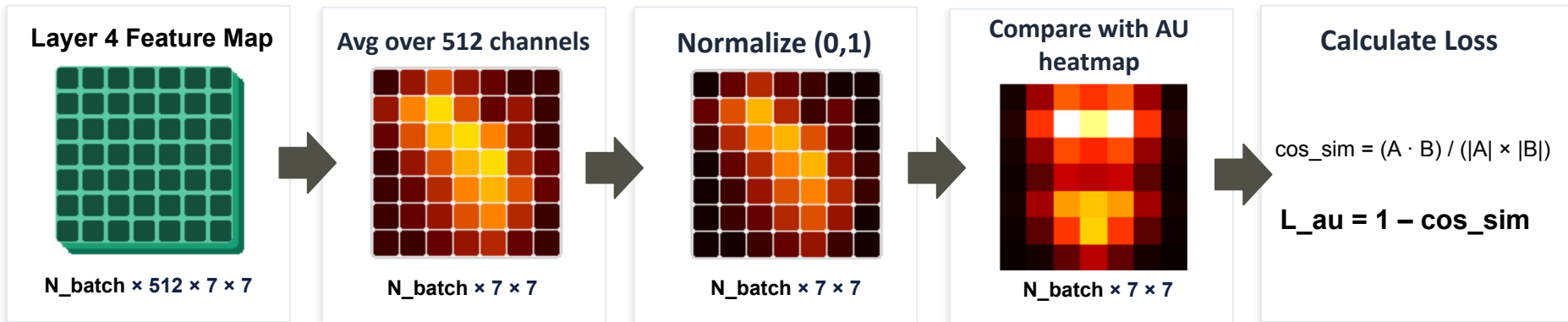


Neutral

No focus

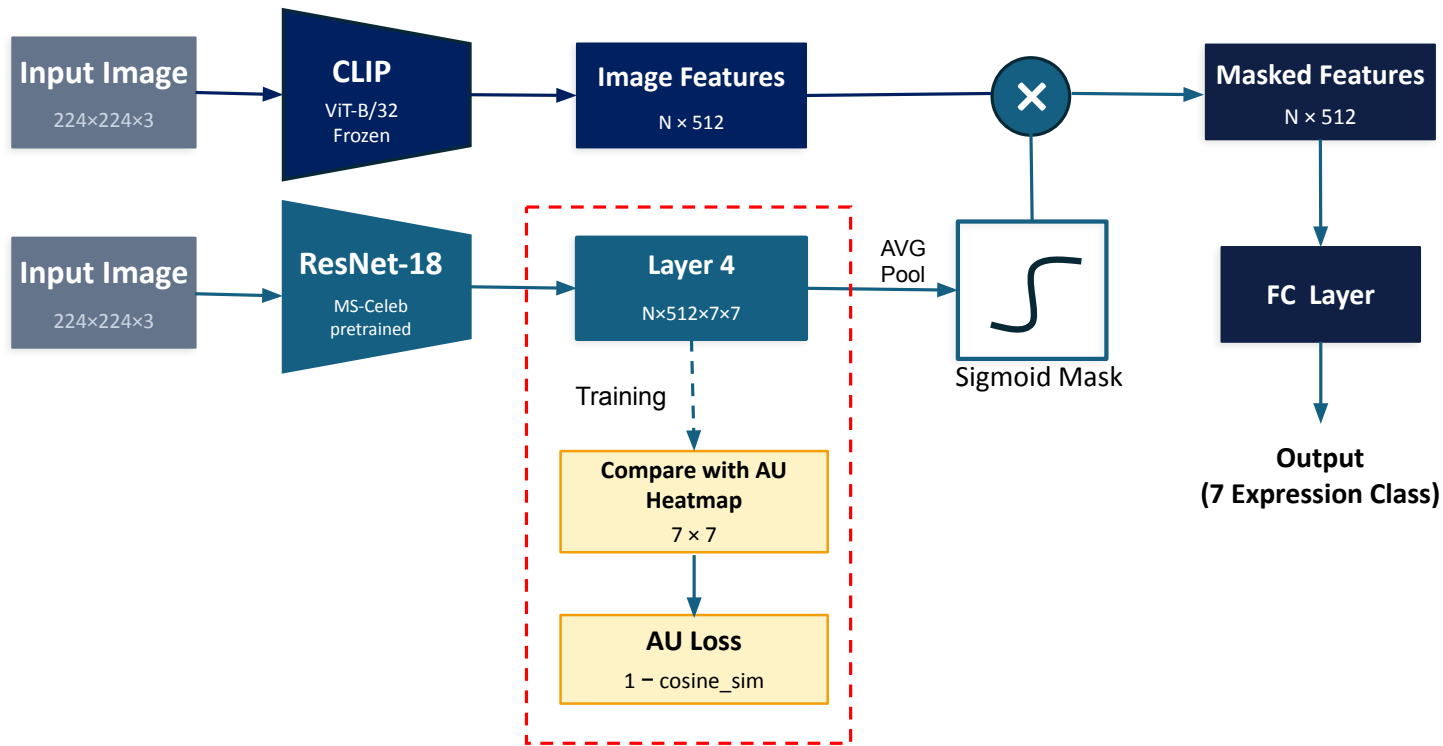
Generalizable-FER

How Does It Work ?

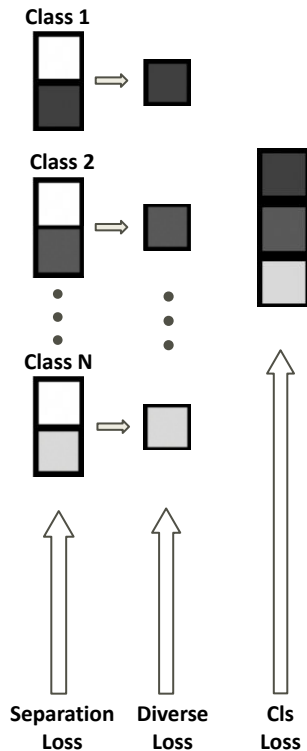


Generalizable-FER

Proposed Pipeline



(Added into our final loss computation)



Generalizable-FER

Improvement

2. Class Imbalance Loss

- We reduce bias toward common classes
- Giving higher weights to class with fewer samples

	Surprise	Fear	Disgust	Happy	Sad	Angry	Neutral
FER+	3172	510	107	7335	3047	2054	9030
RAF-DB	1290	281	717	4772	1982	705	2524

(Training Datasets For Each Classes)

$$W_{class} = \frac{N_{total}}{C \cdot N_c}$$

$$l_{cls} = - \frac{1}{N} \sum_{i=1}^N \mathbf{W}_{class} \cdot \log \frac{e^{\mathbf{W}_{y_i} \tilde{\mathbf{F}}_i}}{\sum_j^L e^{\mathbf{W}_j \tilde{\mathbf{F}}_i}}$$

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3. Total Loss

$$l_{cls} = - \frac{1}{N} \sum_{i=1}^N \mathbf{W}_{class} \cdot \log \frac{e^{\mathbf{W}_{y_i} \tilde{\mathbf{F}}_i}}{\sum_j^L e^{\mathbf{W}_j \tilde{\mathbf{F}}_i}}$$

$$l_{sep} = - \frac{1}{N} \sum_{i=1}^N \left(\log \frac{e^{\bar{\mathbf{F}}_{y_i}^d}}{\sum_j^L e^{\bar{\mathbf{F}}_j^d}} \right)$$

$$l_{div} = 1 - \frac{1}{N_c} \sum_{i=1}^N \sum_{j=1}^L \tilde{\mathbf{F}}_{max_{ij}}$$

$$W_{class} = \frac{N_{total}}{C \cdot N_c}$$

$$\bar{\mathbf{F}} = \tilde{\mathbf{F}} \mathbf{M}_{drop} = \{\tilde{\mathbf{F}}_1 \mathbf{M}_1, \tilde{\mathbf{F}}_2 \mathbf{M}_2, \dots, \tilde{\mathbf{F}}_L \mathbf{M}_L\}.$$

$$\bar{\mathbf{F}}^d = \{max(\tilde{\mathbf{F}}_1 \mathbf{M}_1), max(\tilde{\mathbf{F}}_2 \mathbf{M}_2), \dots, max(\tilde{\mathbf{F}}_L \mathbf{M}_L)\}$$

$$\tilde{\mathbf{F}}_{max} = \{max(\tilde{\mathbf{F}}_1), max(\tilde{\mathbf{F}}_2), \dots, max(\tilde{\mathbf{F}}_L)\}$$

$$l_{total} = l_{cls} + \lambda_1 l_{sep} + \lambda_2 l_{div} + \lambda_3 \mathbf{L}_{AU}$$

Evaluation: **Accuracy** across variety of datasets

Data Augmentation

Motivations:

- Simulate variations in real world data
- Prevents the model from overfitting to the training data
- Helps the training distribution to be more similar to the unseen test data distribution

(Gattermayr, 2025)

Data Augmentation

Baseline augmentation composition:

- Horizontal Flipping,
- Random Erasing,
- Gaussian Noise

Proposed augmentation composition:

- Horizontal Flipping,
- Rotation,
- Gaussian Noise,
- Color Jittering,
- Cropping

(Gattermayr, 2025)

Ablation table (Accuracy)

Index	Method			Training Dataset	Dataset only used for evaluation				Mean
	(1)	(2)	(3)	RAF-DB	FERPlus	AffectNet	SFEW2.0	MMA	
1	✖	✖	✖	88.33%	64.78%	50.62%	42.69%	57.30%	60.74%
2	✔	✖	✖	87.22%	64.52%	49.40%	45.71%	57.18%	60.81%
3	✖	✔	✖	88.48%	63.48%	51.29%	46.40%	56.49%	61.23%
4	✔	✔	✖	88.66%	60.94%	53.10%	45.48%	55.85%	60.81%
5	✖	✖	✔	88.66%	69.52%	49.65%	46.40%	58.57%	62.56%

- (1) AU-guided Mask
- (2) Class Imbalance Loss
- (3) Data Augmentation

AU/CIL: inconsistent improvements, helps some datasets, hurts others
 Augmentation: most consistently improved across all datasets

Ablation table (F1 score)

Index	Method			Training Dataset	Dataset only used for evaluation				Mean
	(1)	(2)	(3)		FERPlus	AffectNet	SFEW2.0	MMA	
1	✗	✗	✗	0.80	0.50	0.41	0.32	0.43	0.49
2	✓	✗	✗	0.78	0.49	0.40	0.33	0.43	0.49
3	✗	✓	✗	0.81	0.51	0.45	0.36	0.44	0.51
4	✓	✓	✗	0.81	0.51	0.47	0.37	0.44	0.52
5	✗	✗	✓	0.80	0.54	0.41	0.44	0.44	0.53

- (1) AU-guided Mask
- (2) Class Imbalance Loss
- (3) Data Augmentation

AU + CIL: Improved across all 5 test datasets
 (Setting the Direction + Providing the Guidance)
 Augmentation: highest mean F1 score

Ablation table (Accuracy)

Index	Method			Training Dataset	Dataset only used for evaluation				Mean
	(1)	(2)	(3)	SFEW2.0	RAF-DB	FERPlus	AffectNet	MMA	
1	✗	✗	✗	47.03%	52.02%	42.85%	29.08%	37.81%	41.76%
2	✓	✗	✗	44.78%	52.90%	40.39%	29.19%	37.72%	41.00%
3	✗	✓	✗	44.55%	48.70%	34.43%	27.11%	37.62%	38.48%
4	✓	✓	✗	44.55%	42.14%	39.97%	37.58%	24.91%	37.83%
5	✗	✗	✓	47.56%	50.72%	39.12%	27.93%	37.34%	40.53%

- (1) AU-guided Mask
- (2) Class Imbalance Loss
- (3) Data Augmentation

All 4 models had lower mean accuracy worse than baseline,
- inconsistent improvements

Ablation table (F1 score)

Index	Improvement			Training Dataset	Dataset only used for evaluation				Mean
	(1)	(2)	(3)	SFEW2.0	RAF-DB	FERPlus	AffectNet	MMA	
1	✗	✗	✗	0.31	0.31	0.26	0.20	0.21	0.26
2	✓	✗	✗	0.30	0.31	0.25	0.21	0.21	0.26
3	✗	✓	✗	0.34	0.34	0.22	0.21	0.22	0.27
4	✓	✓	✗	0.34	0.29	0.22	0.18	0.20	0.25
5	✗	✗	✓	0.32	0.30	0.24	0.20	0.20	0.25

- (1) AU-guided Mask
- (2) Class Imbalance Loss
- (3) Data Augmentation

Most models had a lower mean F1 score, except CIL

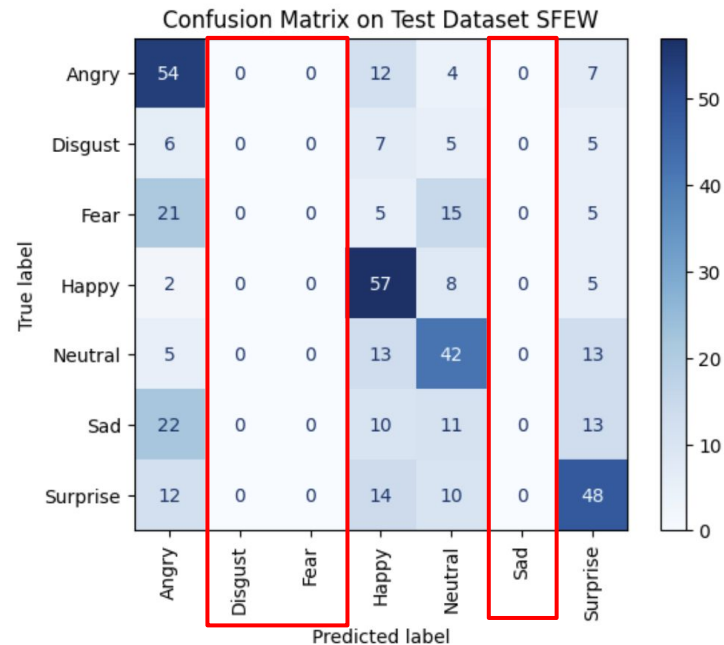
Discussions

Overall model performance:

- Mean accuracy and F1 score improved when trained with RAFDB, but performed worse with SFEW2.0

Observation:

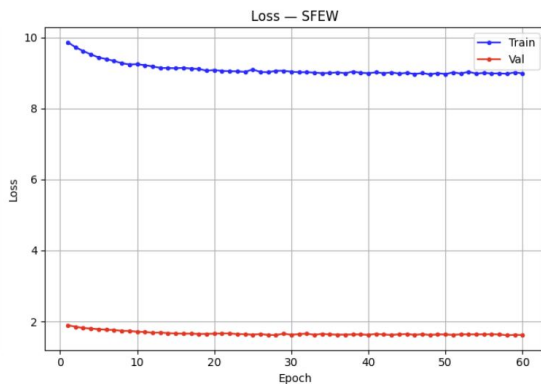
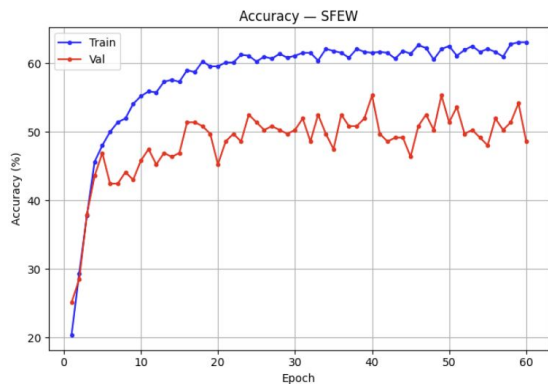
- SFEW2.0 dataset is small, noisy and imbalanced
- Baseline model does not predict Disgust, Fear nor Sad when trained with SFEW2.0



Discussion

Reasons for poorer performance on SFEW2.0:

- Baseline model has signs of underfitting → already struggles with basic emotion classification



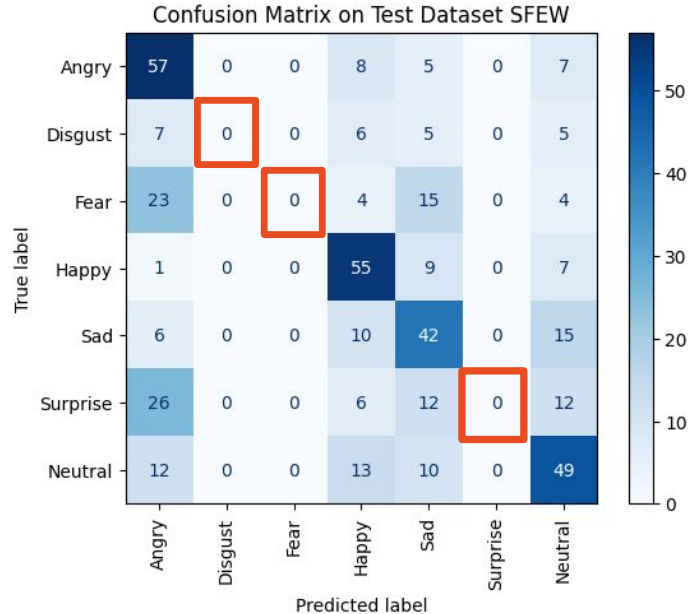
- Proposed improvements may further complicate optimization → not only learn expression representations + new constraints

Discussions (Class Imbalance Loss)

- Accuracy decreased slightly due to the trade-off for a more balanced model
- It sacrifices accuracy of majority classes to increase accuracy for minority classes

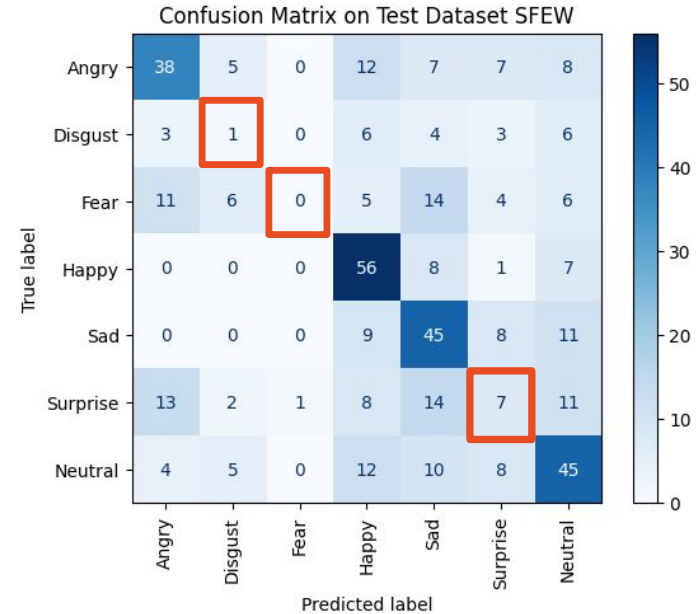
Discussions (Class Imbalance Loss)

BEFORE



Acc	Macro f1
47.03	0.31

AFTER



Acc	Macro f1
44.50	0.34

Discussions (AU mask)

- This method hurts our model performance because the spatial heatmap **collapse** the 512 channels into one 7x7 map and **loses channel-level information**
- The model cannot identify which specific channels activate at AU-relevant regions like brows or mouth.
- This results in inconsistent performance across datasets

Discussions (further improvement)

Possible further improvements to explore:

- Manual augmentation of less represented classes
- Use an **attention mechanism** to weight each of the 512 channels based on how relevant they are to the AU heatmap instead of the cosine similarity per channel.
- Adding macro f1 as one of our evaluation metrics to show overall performance across all classes instead of only using Accuracy

Thank You